Sihao Liu

Research Scientist, Architecture Research Group, NVIDIA Research

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Research Interests

- Programmable spatial accelerators, hardware/software co-design, compilers, and on-chip interconnects.
- Agentic chip architecture and electronic design automation; independently reviewed and executable engineering workflows.

Education

2026 • **Ph.D. in Computer Science** — University of California, Los Angeles

Degree awarded September 2026. Advisor: Tony Nowatzki.
Dissertation: Loom: Full-Stack Compilation and Automated Exploration of Heterogeneous Multi-Core Spatial Accelerator Systems.

2021 • **M.S. in Computer Science** — University of California, Los Angeles

2019 • **B.Eng. in Electrical Engineering** — Xi'an Jiaotong University

Professional Experience

09/2026–present • **Research Scientist — NVIDIA Research**
Architecture Research Group. Research on next-generation chip architectures and agentic systems for hardware design.

06/2022–10/2023 • **Hardware Engineering Intern — SiFive**
Memory management unit ownership; load/store and cache subsystem design and verification. Full-time summer and part-time academic-year appointments.

Summer 2021 • **CPU Microarchitecture Intern — Futurewei Technologies**
RISC-V vector CPU research and FPGA platform development.

Academic Experience

2019–2026 • **Doctoral Researcher** — UCLA PolyArch Research Group

2018 • **Undergraduate Researcher (summer)** — UCLA PolyArch Research Group

2017 • **Visiting Scholar (summer)** — University of Lausanne
Neuroheuristic Research Group. Supervisor: Alessandro E. P. Villa.

Research Projects and Open Source

2026 • **Loom**. A compiler and exploration framework for heterogeneous multi-core spatial accelerators, connecting a shared hardware model, compilation, simulation, and reproducible evaluation.

Open source • **Humanize / Humanfia**. I created Humanize 1 at UCLA: an implementation and independent-review loop for coding agents. I am a founding member of Humanfia, where we extend this approach to longer engineering tasks.

2025–2026 • **OverNoC / BenchNoC**. Characterization of hardened FPGA networks-on-chip and hierarchical network overlays for multi-chiplet reconfigurable systems.

2020–2022 • **DSAGEN / OverGen**. Automated generation of programmable spatial architectures and FPGA overlays, bringing compilation, hardware implementation, and design-space exploration into one flow.

Research Impact

913 citations; h-index 11; i10-index 11. Google Scholar snapshot, September 29, 2026.

Conference Publications

* Equal contribution. Preprints and the dissertation are listed separately.

FCCM 2025
- H. Ke, **S. Liu**, L. Guo, Z. He, L. Song, S. Basalama, Y. Chi, T. Nowatzki, J. Cong. NoH: NoC Compilation in High-Level Synthesis.

MICRO 2022
- **S. Liu**[*], J. Weng[*], D. Kupsh, A. Sohrabizadeh, Z. Wang, L. Guo, J. Liu, M. Zhulin, R. Mani, L. Zhang, J. Cong, T. Nowatzki. OverGen: Improving FPGA Usability through Domain-Specific Overlay Generation. **Best Paper Runner-up**.

HPCA 2022
- Z. Wang, J. Weng, **S. Liu**, T. Nowatzki. Near-Stream Computing: General and Transparent Near-Cache Acceleration.

ISCA 2021
- V. Dadu, **S. Liu**, T. Nowatzki. PolyGraph: Exposing the Value of Flexibility for Graph Processing Accelerators. **IEEE Micro Top Picks 2022**.

ISCA 2020
- J. Weng[*], **S. Liu**[*], V. Dadu, Z. Wang, P. Shah, T. Nowatzki. DSAGEN: Synthesizing Programmable Spatial Accelerators. **IEEE Micro Top Picks 2021, Honorable Mention**.

HPCA 2020
- J. Weng, **S. Liu**, Z. Wang, V. Dadu, T. Nowatzki. A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms.

MICRO 2019
- V. Dadu, J. Weng, **S. Liu**, T. Nowatzki. Towards General Purpose Acceleration by Exploiting Common Data-Dependence Forms. **IEEE Micro Top Picks 2020**.

MICRO 2019
- A. Sharifian, R. Hojabr, N. Rahimi, **S. Liu**, A. Guha, T. Nowatzki, A. Shriraman. μIR: An Intermediate Representation for Transforming and Optimizing the Microarchitecture of Application Accelerators.

IUS 2017
- S. Zhang, S. Shang, Y. Han, R. Xu, **S. Liu**, L. Zhang, S. Wu, G. Niu, S. Wang, M. Wan. Monitoring of Microwave Ablation in Porcine Liver Using Ultrasonic Nakagami Imaging.

Journal Publications

Micro 2022
- J. Weng, **S. Liu**, D. Kupsh, T. Nowatzki. Unifying Spatial Accelerator Compilation With Idiomatic and Modular Transformations.
  IEEE Micro, 42(5), 59-69.

Micro 2022
- V. Dadu, **S. Liu**, T. Nowatzki. Systematically Understanding Graph Accelerator Dimensions and the Value of Hardware Flexibility.
  IEEE Micro, 42(4), 87-96.

Micro 2020
- V. Dadu, J. Weng, **S. Liu**, T. Nowatzki. Towards General-Purpose Acceleration: Finding Structure in Irregularity.
  IEEE Micro, 40(3), 37-46.

CAL 2019
- J. Weng, **S. Liu**, V. Dadu, T. Nowatzki. DAEGEN: A Modular Compiler for Exploring Decoupled Spatial Accelerators.
  IEEE Computer Architecture Letters, 18(2), 161-165.

TMI 2018
- S. Zhang, S. Shang, Y. Han, C. Gu, S. Wu, **S. Liu**, G. Niu, A. Bouakaz, M. Wan. Ex Vivo and In Vivo Monitoring and Characterization of Thermal Lesions by High-Intensity Focused Ultrasound and Microwave Ablation Using Ultrasonic Nakagami Imaging.
  IEEE Transactions on Medical Imaging, 37(7), 1701-1710.

IJH 2018
- S. Zhang, R. Xu, S. Shang, Y. Han, **S. Liu**, T. Xu, C. Gu, X. Zhu, G. Niu, M. Wan. In Vivo Monitoring of Microwave Ablation in a Porcine Model Using Ultrasonic Differential Attenuation Coefficient Intercept Imaging.
  International Journal of Hyperthermia, 34(8), 1157-1170.

Dissertation, Preprints, Workshops, and Book Chapters

Thesis 2026
- **S. Liu**. Loom: Full-Stack Compilation and Automated Exploration of Heterogeneous Multi-Core Spatial Accelerator Systems.
  Ph.D. dissertation, University of California, Los Angeles.

arXiv 2026
- C. Li, Z. Zhang, M. Li, Y. Pang, T. Wu, M. Li, Y. Liu, H. Guo, M. Chen, C. He, K. Ma, W. Xu, S. Han, **S. Liu**, L. Zhu. Humanize MLE Agents: Multi-Agent Session Alternation Beyond Single-Agent Limits.
  arXiv:2610.13494.

arXiv 2026
- **S. Liu**, L. Zhu, Z. Zhang, D. Zou, Z. Zhang, C. Li, S. Bian, S. Han, T. Nowatzki. Humanize: Judgement Engineering for Agentic Coding.
  arXiv:2610.08900.

arXiv 2025
- **S. Liu**, J. Ke, T. Nowatzki, J. Cong. Demystifying FPGA Hard NoC Performance.
  arXiv:2503.10861.

WDDSA 2022
- J. Weng, **S. Liu**, D. Kupsh, T. Nowatzki. A Full-Stack Infrastructure for Automating Spatial Architecture Research.
  Workshop on Democratizing Domain-Specific Accelerators (WDDSA), co-located with MICRO.

LATTE 2021
- J. Weng, V. Dadu, **S. Liu**, T. Nowatzki. Generality is the Key Dimension in Accelerator Design.
  Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE), co-located with ASPLOS.

Chapter 2021
- Q. Zheng, **S. Liu**, A. E. P. Villa, A. Lintas. Unsupervised Analysis of EEG Signals Reveals Common Personality Traits During an Iterated Ultimatum Game.
  Advances in Cognitive Neurodynamics (VII), Springer, pp. 275-276.

Honors and Awards

2022
- **MICRO Best Paper Runner-up**. OverGen.

2022
- **IEEE Micro Top Picks**. PolyGraph.

2021
- **IEEE Micro Top Picks, Honorable Mention**. DSAGEN.

2020
- **IEEE Micro Top Picks**. SPU.

2019
- **Dean's Fellowship and University Fellowship**. UCLA.

Academic Service

2021
- Artifact Evaluation Committee, HPCA 2021.

2021–2026
- Journal peer reviewer: ACM TACO, ACM TODAES, ACM TRETS, ACM TECS, IEEE TC, IEEE TCAD, IEEE TVLSI, IEEE CAL, IEEE JETCAS, Journal of Systems Architecture, Journal of Signal Processing Systems.

Technical Skills

- **Programming:** C, C++, Rust, Python, Scala, MATLAB.
- **Hardware:** Chisel, Verilog, SystemVerilog; RTL design and verification; FPGA prototyping.
- **Compilers and simulation:** LLVM, MLIR, CIRCT, gem5, Chipyard, Rocket Chip, FireSim.
- **EDA tools:** Synopsys VCS, Design Compiler, IC Compiler; Verilator; AMD Vivado and Vitis.

Earlier Honors and Competitions

2017
- Finalist, Interdisciplinary Contest in Modeling.

2017
- First Prize, China Big Data Modeling Contest.

2016
- First Prize, Data Modeling Contest, XJTU.

2016
- First Prize, Mathematical Modeling Contest, XJTU.

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